CSE STUDY HUB

Digital Forensics & Incident Response

UNIT 1 · Fundamentals of Digital Forensics

What this unit is about

This unit builds understanding of **Fundamentals of Digital Forensics**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Digital forensics: definition, objectives and investigation lifecycle

2. Branches: computer, mobile, network, cloud, database and memory forensics

3. Cybercrime examples: phishing, ransomware, identity theft, cyberstalking, DoS and spoofing

4. Digital evidence: files, logs, emails, metadata, browser artefacts and network records

5. Evidence properties: relevance, authenticity, integrity, reliability and admissibility

6. Volatile vs non-volatile evidence; order of volatility

7. Live forensics vs dead-box (dead) forensics

8. Chain of custody: collection, sealing, transfer, storage and examination records

9. Forensic process and legal/ethical considerations; NIST guidance and ISO/IEC 27037

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 1 NOTES · PART 1

Digital forensics: definition, objectives and investigation lifecycle

Start with a precise definition of Digital forensics: definition, objectives and investigation lifecycle. Explain its purpose, the problem it solves, its main components and where it is used. In an exam answer, add one practical example and distinguish it from the nearest related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Branches: computer, mobile, network, cloud, database and memory forensics

Cloud computing provides on-demand shared computing resources over a network. IaaS supplies infrastructure, PaaS supplies a development platform, and SaaS supplies a ready-to-use application.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Cybercrime examples: phishing, ransomware, identity theft, cyberstalking, DoS and spoofing

Cybercrime examples: phishing, ransomware, identity theft, cyberstalking, DoS and spoofing is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Fundamentals of Digital Forensics, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 1 NOTES · PART 2

Digital evidence: files, logs, emails, metadata, browser artefacts and network records

A professional email has a specific subject, suitable greeting, purpose-first opening, concise body, clear action/deadline, polite close and signature. Check tone, recipients and attachments before sending.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Evidence properties: relevance, authenticity, integrity, reliability and admissibility

Evidence properties: relevance, authenticity, integrity, reliability and admissibility is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Volatile vs non-volatile evidence; order of volatility

Volatile evidence can disappear when power is lost or a system changes. RAM, running processes, active network connections and logged-in sessions should generally be considered before persistent disk data.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Fundamentals of Digital Forensics, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 1 NOTES · PART 3

Live forensics vs dead-box (dead) forensics

Compare the alternatives in Live forensics vs dead-box (dead) forensics using the same criteria: purpose, working method, performance, cost/complexity, security or reliability, and suitable use case. End with a short conclusion stating when each option is preferable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Chain of custody: collection, sealing, transfer, storage and examination records

Chain of custody is the time-stamped record of who collected, handled, transferred, stored and examined evidence. Each action should identify the person, date/time, purpose and seal/identifier.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Forensic process and legal/ethical considerations; NIST guidance and ISO/IEC 27037

Present Forensic process and legal/ethical considerations; NIST guidance and ISO/IEC 27037 as an ordered workflow. State the preconditions, list the major steps, explain what each step produces, and mention how errors are detected or handled. Use a small worked example or flowchart in the exam.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Fundamentals of Digital Forensics, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Digital forensics: definition, objectives and investigation lifecycle** — know the definition, core working and one example.

• **Branches: computer, mobile, network, cloud, database and memory forensics** — know the definition, core working and one example.

• **Cybercrime examples: phishing, ransomware, identity theft, cyberstalking, DoS and spoofing** — know the definition, core working and one example.

• **Digital evidence: files, logs, emails, metadata, browser artefacts and network records** — know the definition, core working and one example.

• **Evidence properties: relevance, authenticity, integrity, reliability and admissibility** — know the definition, core working and one example.

• **Volatile vs non-volatile evidence; order of volatility** — know the definition, core working and one example.

• **Live forensics vs dead-box (dead) forensics** — know the definition, core working and one example.

• **Chain of custody: collection, sealing, transfer, storage and examination records** — know the definition, core working and one example.

• **Forensic process and legal/ethical considerations; NIST guidance and ISO/IEC 27037** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Fundamentals of Digital Forensics. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Digital forensics: definition, objectives and investigation lifecycle and explain its importance.

Q2. Explain the working/steps involved in Branches: computer, mobile, network, cloud, database and memory forensics.

Q3. Differentiate between two related concepts from Fundamentals of Digital Forensics.

Q4. Give one practical application of Forensic process and legal/ethical considerations; NIST guidance and ISO/IEC 27037 and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Fundamentals of Digital Forensics.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.